¿Relaciones Causales en Redes Neuronales o Inferencias Inductivas?

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Febrero, 2026

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Facultad de Ciencias UNAM

AGILE DESIGN STUDIO

INTERNATIONAL CONFERENCE ON PHILOSOPHY OF COMPUTING
UNAM MEXICO CITY 2025

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¿Hemos logrado substraer relaciones causales en los modelos más recientes de redes neuronales o LLMs? ¿Qué nos demuestra la posibilidad de inferir causalidades que se observan en nuestra realidad observable a partir de estos modelos?

Importancia de la pregunta

No solo existe una amplia gama de toma decisiones basadas en el resultado de estos modelos, existes diversos programas enfocados en usar modelos para encontrar descubrimientos científicos con técnicas de aprendizaje automático (ej. [1, 2, 3, 4]). Si las inferencias causales constituyen nuestra manera de hacer conocimiento, tenemos que demostrar que dichos modelos son capaces de sustraerlas.

Motivaciones Epistémicas

“This becomes a critical issue because science represents the set of tools humanity uses for exploring and understanding ourselves, our universe, and our place in the web of life here on Earth.We need to be able to trust our tools to help us solve problems in the biosphere without generating side effects worse then the problems, themselves."(Judith Rosen, [5])

Explicabilidad

La causalidad es uno de las cualidades que buscamos para intentar mejorar la explicabilidad de los modelos de aprendizaje profundo. Resaltado por [6] buscamos lo siguiente:

- **Equidad**: garantizar que las predicciones sean imparciales y no discriminen, de forma implícita o explícita, a grupos subrepresentados.
- **Privacidad**: garantizar que la información sensible en los datos esté protegida.
- **Confiabilidad o Robustez**: garantizar que pequeños cambios en la entrada no provoquen grandes cambios en la predicción.
- **Causalidad**: asegurar que solo se identifiquen relaciones causales.
- **Confianza**: para las personas es más fácil confiar en un sistema que explica sus decisiones que en una caja negra.

Relacion de Modelado

1. Las relaciones existen en el mundo real
2. Tales relaciones pueden ser descubiertas por la mente. Este descubrimiento no es directo, como el de las cualidades, sino que se imputa a la realidad desde el constructo mental.
3. Diremos que una cualidad de un NS es observable cuando tal cualidad sea perceptible. Diremos que una relación producible entre dos observables es un **vínculo** (linkage). [7]

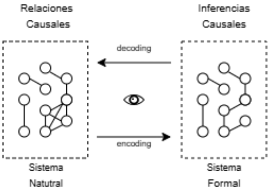


Figure: Relación de modelado entre la realidad, los modelos y las inferencias causales.

Antecedentes

Existe una amplia gama de literatura en el aprendizaje causal, e intentar replicarlo en modelos de aprendizaje profundo es un área de investigación activa. Sin embargo, la mayoría de los enfoques actuales se centran en la inferencia causal a partir de datos estructurados, como gráficos causales (DAGs).

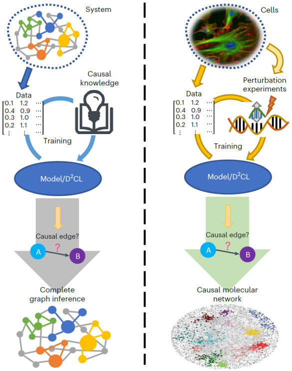
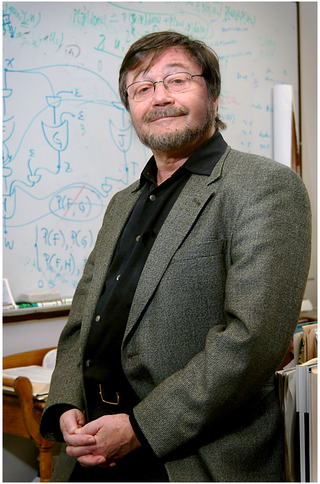


Figure: Deep learning of causal structures in high dimensions under data limitations [8]

Judea Pearl

Uno de los mayores promotores de la idea de que las redes neuronales no pueden inferir relaciones causales es Judea Pearl, quien ha argumentado que los modelos de aprendizaje profundo, como los LLMs, carecen de la capacidad de representar y razonar sobre relaciones causales. Según Pearl, estos modelos se basan en correlaciones estadísticas y no pueden distinguir entre correlación y causalidad. [9, 10, 11]



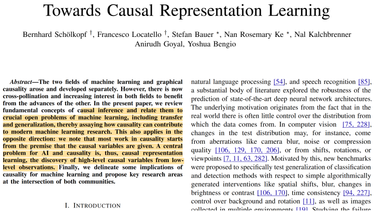
The Three Layer Causal Hierarchy

Level (Symbol)	Typical Activity	Typical Questions	Examples
1. Association $P(y\|x)$	Seeing	What is? How would seeing X change my belief in Y?	What does a symptom tell me about a disease? What does a survey tell us about the election results?
2. Intervention $P(y\|do(x), z)$	Doing Intervening	What if? What if I do X?	What if I take aspirin, will my headache be cured? What if we ban cigarettes?
3. Counterfactuals $P(y_x\|x', y')$	Imagining, Retrospection	Why? Was it X that caused Y? What if I had acted differently?	Was it the aspirin that stopped my headache? Would Kennedy be alive had Oswald not shot him? What if I had not been smoking the past 2 years?

Figure 1: The Causal Hierarchy. Questions at level i can only be answered if information from level i or higher is available.

Figure: La jerarquía causal de Pearl, que ilustra los diferentes niveles de inferencia causal y cómo los modelos de aprendizaje profundo se sitúan en el nivel más bajo, el de la asociación. Aunque desarrollo algunos métodos fusionando contrafácticos con teoría de grafos

Yoshua Bengio



Towards Causal Representation Learning

Bernhard Schölkopf †, Francesco Locatello †, Stefan Bauer *, Nan Rosemary Ke *, Nal Kalchbrenner
Anirudh Goyal, Yoshua Bengio

Abstract—**The two fields of machine learning and graphical causality arose and developed separately. However, there is now cross-pollination and increasing interest in both fields to benefit from the advances of the other. In the present paper, we review fundamental concepts of causal inference and relate them to crucial open problems of machine learning, including transfer and generalization, thereby assaying how causality can contribute to modern machine learning research. This also applies in the opposite direction: we note that most work in causality starts from the premise that the causal variables are given. A central problem for AI and causality is, thus, causal representation learning, the discovery of high-level causal variables from low-level observations. Finally, we delineate some implications of causality for machine learning and propose key research areas at the intersection of both communities.**

I. Introduction

natural language processing [54], and speech recognition [85], a substantial body of literature explored the robustness of the prediction of state-of-the-art deep neural network architectures. The underlying motivation originates from the fact that in the real world there is often little control over the distribution from which the data comes from. In computer vision [75, 228], changes in the test distribution may, for instance, come from aberrations like camera blur, noise or compression quality [106, 129, 170, 206], or from shifts, rotations, or viewpoints [7, 11, 63, 282]. Motivated by this, new benchmarks were proposed to specifically test generalization of classification and detection methods with respect to simple algorithmically generated interventions like spatial shifts, blur, changes in brightness or contrast [106, 170], time consistency [94, 227], control over background and rotation [11], as well as images

Figure: Toward causal representation learning[12]

Otras investigaciones se centran en modelar o encontrar inferencias causales en LLM's, modelando los datos a traves de *causal embeddings*.

Causal Discovery and Inference through Next-Token Prediction

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Abstract

Deep neural networks have been criticized as fundamentally *statistical* systems that fail to capture causal structure and perform causal reasoning. Here we demonstrate that a GPT-style transformer trained for next-token prediction can simultaneously discover instances of linear Gaussian structural causal models (SCMs) and learn to answer counterfactual queries about those SCMs. First, we show that the network generalizes to counterfactual queries about SCMs for which it has seen interventional data but not any examples of counterfactual inference. The network must, thus, have successfully composed discovered causal structures with a learned counterfactual inference algorithm. Second, we decode the implicit "mental" SCM from the network's residual stream activations and manipulate it using gradient descent with predictable effects on the network's output. Our results suggest that statistical prediction may be sufficient to drive the emergence of internal causal models and causal inference capacities in deep neural networks.

Estado del Arte

Existe un programa llamado *Topological Deep Learning* (Topological Deep Learning is the New Frontier for Relational Learning)[13] que explora invariantes geométricas y topológicas en los modelos de redes neuronales. La idea central es encontrar diferentes maneras de obtener resultados con aprendizaje profundo y datos relacionales. Para entender y lograr capturar las relaciones subyacentes a los datos.

Position: Topological Deep Learning is the New Frontier for Relational Learning

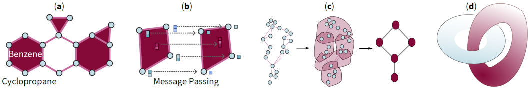
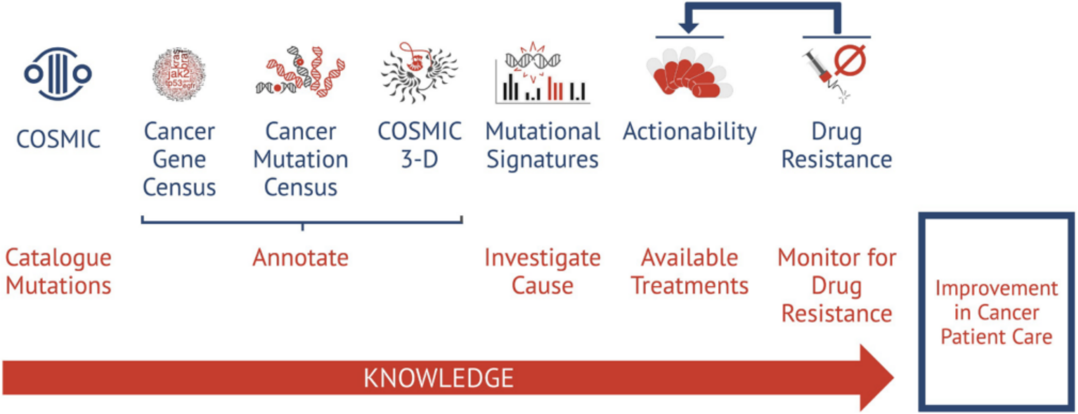


Figure 1. Topological spaces enrich deep learning methods from a variety of perspectives. (a): A topological space, modeled as a cell complex, enables a flexible molecular representation. This representation can improve the performance of a deep learning model supported on this space (Bodnar et al., 2021a). (b): Topological neural networks enable the processing of data, for example via higher-order message-passing schemes on a topological space. These networks find a wide range of applications, from computer graphics to drug discovery (Hajij et al., 2023b). (c): Topological spaces allow hierarchical representations of the underlying data that naturally correspond to pooling operations in deep learning (Hajij et al., 2023b). (d): The topological characteristics of the underlying data are crucial when selecting a neural network architecture. Data from a knotted structure in $\mathbb{R}^3$, such as the one shown, cannot be embedded in $\mathbb{R}^2$ with a single layer multilayer perceptron (MLP) from $\mathbb{R}^3$ to $\mathbb{R}^2$ (Olah, 2014).

Figure: Ejemplo en células.[13]

- La topología del espacio de datos subyacente determina qué arquitecturas de redes neuronales son viables
- Los dominios topológicos permiten modelar datos que contienen interacciones de múltiples niveles (relaciones de orden superior / **complejidad**)
- El aprendizaje profundo topológico captura regularidades propias de las variedades (**explicabilidad** en LLM’s).
- Detecta las equivarianzas topológicas presentes en los datos (**inferencias causales**).
- Integra características topológicas inherentes a datos relacionales. (**interpretabilidad**).

COSMIC Cancer Database



Relaciones en la Ciencias

COSMIC es una base de datos de mutaciones genéticas en cáncer, que se ha utilizado para entrenar modelos de redes neuronales con el fin de identificar genes impulsores del cáncer.
Artículos recientes:

- Trans-Driver: A Deep Learning Approach for Cancer Driver Gene Discovery With Multi-Omics Data [14]
- Explainable Multilayer **Graph Neural Network** for cancer gene prediction [15]
- DGMP: Identifying Cancer Driver Genes by Jointing DGCN and MLP from Multi-omics Genomic Data (**Directed graph convolutional network**, Multilayer perceptron) [16]

Gráficas de Conocimiento

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SCICERO: A deep learning and NLP approach for generating scientific knowledge graphs in the computer science domain

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ARTICLE INFO

Article history:
Received 3 August 2022
Received in revised form 13 September 2022
Accepted 25 September 2022
Available online 5 October 2022

Keywords:
Knowledge graph
Scholarly domain
Scientific facts
Artificial intelligence

ABSTRACT

Science communication has a number of bottlenecks that include the rising number of published research papers and its non-machine-accessible and document-based paradigm, which makes the exploration, reading, and reuse of research outcomes rather inefficient. Recently, Knowledge Graphs (KG), i.e., semantic interlinked networks of entities, have been proposed as a new core technology to describe and curate scholarly information with the goal to make it machine readable and understandable. However, the main drawback of the use of such a technology is that researchers are asked to manually annotate their research papers and add their contributions within the KGs. To address this problem, in this paper we propose SCICERO, a novel KG generation approach that takes in input text from research articles and generates a KG of research entities. SCICERO uses Natural Language Processing techniques to parse the content of scientific papers to discover entities and relationships, exploits state-of-the-art Deep Learning Transformer models to make sense and validate extracted information, and uses Semantic Web best practices to formally represent the extracted entities and relationships, making the written content of research papers machine-actionable. SCICERO has been tested on a dataset of 6.7M papers about Computer Science generating a KG of about 10M entities. It has been evaluated on a manually generated gold standard of 3,600 triples that cover three Computer Science subdomains (Information Retrieval, Natural Language Processing, and Machine Learning) obtaining remarkable results.

Figure: Ejemplo de la construcción de un grafo de conocimiento científico, que representa relaciones entre conceptos, descubrimientos y publicaciones en un campo específico.[17]

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Unmasking Hidden Fraud Patterns: An Explainable Graph Attention Network Approach for Highly Imbalanced Financial Datasets(Oguntola, 2026)

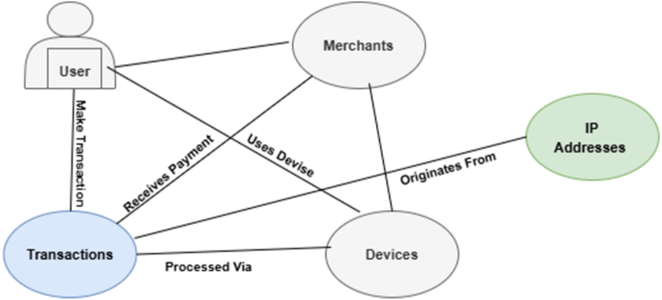

Figure: Ejemplo de una red neuronal gráfica, que detecta fraudes fiscales la ventaja comparativa hace que tenga un desempeño superior a otras arquitecturas de redes neuronales además de permitirnos analizar relaciones complejas entre entidades.[18]

Manifolds

Variedades(Espacios curvos multi dimensionales $\mathbb{R}^n$). Este tipo de aproximaciones son el trabajo de ingeniería inversa con respecto al ordenamiento de los datos. A comparación de las arquitecturas tipo DAG, esta metodología se vuelve más top-down, ya que se enfoca en la representación de los datos y no en la construcción de una arquitectura específica. Se apoya en la idea de que “Las distancias geodésicas en las variedades de representación de los LLM son significativas” [19].

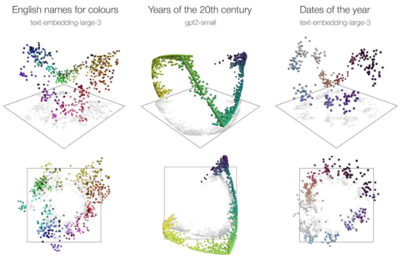

Figure 1: Representation manifolds in large language models: colours, years and dates. The first and third example show text embeddings obtained from OpenAI's `text-embedding-large-3` model from prompts relating to English names for colours and dates of the year, respectivly. The second example shows token activations from layer 7 of `GPT2-small`, which were studied in Engels et al. (2025). The token activations were processed via an SAE to extract a feature corresponding to years of the twentieth century as in Engels et al. (2025), and normalized to have norm one. For each example, we perform principal component analysis (PCA) to reduce the dimension to three and display the resulting point clouds from two perspectives. The embeddings of English names for colours are displayed in their respective colour value. Years are coloured from blue (1900) through green to yellow (1999), and dates are coloured from white (1st Janurary) through blue to black (1st July) through red and back to white.

References I

- Anthony JG Hey, Stewart Tansley, Kristin Michele Tolle, et al.
  The fourth paradigm: data-intensive scientific discovery, volume 1.
  Microsoft research Redmond, WA, 2009.
- Hanchen Wang, Tianfan Fu, Yuanqi Du, Wenhao Gao, Kexin Huang, Ziming Liu, Payal Chandak, Shengchao Liu, Peter Van Katwyk, Andreea Deac, et al.
  Scientific discovery in the age of artificial intelligence.
  Nature, 620(7972):47–60, 2023.
- Yannis Ioannidis.
  The 5th paradigm: Ai-driven scientific discovery.
  Communications of the ACM, 67(12):5–5, 2024.

References II

- Stefano Bianchini, Moritz Müller, and Pierre Pelletier.
  Deep learning in science.
  arXiv preprint arXiv:2009.01575, 2020.
- Robert Rosen (auth.).
  Anticipatory Systems: Philosophical, Mathematical, and Methodological Foundations.
  IFSR International Series in Systems Science and Systems Engineering №1. Springer, 2 edition, 2012.
- Finale Doshi-Velez and Been Kim.
  Towards a rigorous science of interpretable machine learning.
  arXiv preprint arXiv:1702.08608, 2017.

References III

Enrique F. Soto-Astorga.
Codificaciones y la relación de modelado, 2025.
Obsidian Publish.

Kai Lagemann, Christian Lagemann, Bernd Taschler, and Sach Mukherjee.
Deep learning of causal structures in high dimensions under data limitations.
Nature Machine Intelligence, 5(11):1306–1316, 2023.

Judea Pearl.
Causality: Models, Reasoning, and Inference.
Cambridge University Press, 2000.

References IV

Judea Pearl and Dana Mackenzie.
The book of why: the new science of cause and effect.
Basic books, 2018.

Judea Pearl.
Theoretical impediments to machine learning with seven sparks from the causal revolution.
arXiv preprint arXiv:1801.04016, 2018.

Bernhard Schölkopf, Francesco Locatello, Stefan Bauer, Nan Rosemary Ke, Nal Kalchbrenner, Anirudh Goyal, and Yoshua Bengio.
Toward causal representation learning.
Proceedings of the IEEE, 109(5):612–634, 2021.

References V

- Theodore Papamarkou, Tolga Birdal, Michael Bronstein, Gunnar Carlsson, Justin Curry, Yue Gao, Mustafa Hajij, Roland Kwitt, Pietro Lio, Paolo Di Lorenzo, et al.
  Position: Topological deep learning is the new frontier for relational learning.
  Proceedings of machine learning research, 235:39529, 2024.
- Hai Yang, Zhenbei Yang, Lei Zhang, Yijing Yang, Dan Zhou, Dongdong Li, Jing Zhang, and Zhe Wang.
  Trans-driver: a deep learning approach for cancer driver gene discovery with multi-omics data.
  IEEE Transactions on Computational Biology and Bioinformatics, 2025.

References VI

- Michail Chatzianastasis, Michalis Vazirgiannis, and Zijun Zhang.
  Explainable multilayer graph neural network for cancer gene prediction.
  Bioinformatics, 39(11):btad643, 2023.
- Shao-Wu Zhang, Jing-Yu Xu, and Tong Zhang.
  Dgmp: identifying cancer driver genes by jointing dgcn and mlp from multi-omics genomic data.
  Genomics, Proteomics & Bioinformatics, 20(5):928–938, 2022.

References VII

Danilo Dessí, Francesco Osborne, Diego Reforgiato Recupero, Davide Buscaldi, and Enrico Motta.
Scicero: A deep learning and nlp approach for generating scientific knowledge graphs in the computer science domain.
Knowledge-Based Systems, 258:109945, 2022.

Adewale U. Oguntola, Adesola G. Adedayo, Olowookere A. AbdulQoyum, Adesope A. Adetoro, Adebukola M. Madehin, and Ummulkhoyr A. Shittu.
Unmasking Hidden Fraud Patterns: An Explainable Graph Attention Network Approach for Highly Imbalanced Financial Datasets.
PhD thesis, Abiola Ajimobi Technical University, Ibadan, Oyo State, Nigeria, Department of Computer Science, Abiola Ajimobi Technical University, Ibadan, Oyo State, Nigeria, 2026.

References VIII

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Alexander Modell, Patrick Rubin-Delanchy, and Nick Whiteley. The origins of representation manifolds in large language models. *arXiv preprint arXiv:2505.18235*, 2025.